

Теорема Джона и объёмные отношения

1 сентября 2026 года

Every convex body is squeezed between an ellipsoid and its n -fold dilate, and for symmetric bodies the factor drops to $\sqrt{n}$. This is the first result in which the dimension enters the answer explicitly; asymptotic theory starts here.

Definition 1. The **John ellipsoid** of a convex body K is the ellipsoid of maximal volume contained in K ; the **volume ratio** is $\text{vr}(K) = (|K|/|\mathcal{E}|)^{1/n}$.

Theorem 1 (John 1948). The maximal-volume ellipsoid exists and is unique. If it equals B_2^n , then there are contact points $u_i \in \partial K \cap S^{n-1}$ and weights $c_i > 0$ with $\sum_i c_i u_i \otimes u_i = I_n$ and $\sum_i c_i u_i = 0$; consequently $K \subseteq nB_2^n$, and $K \subseteq \sqrt{n}B_2^n$ when $K = -K$.

-
1. Prove that the maximal-volume ellipsoid inside a convex body exists and is unique. (*Hint: Hausdorff compactness together with the strict concavity of $\det^{1/n}$ on positive definite matrices.*)
 2. Prove the easy half of John's theorem: if $B_2^n \subseteq K$ and the contact points admit weights with $\sum c_i u_i \otimes u_i = I$ and $\sum c_i u_i = 0$, then B_2^n is the John ellipsoid of K . Verify the trace identity $\sum_i c_i = n$.
 3. Deduce $K \subseteq nB_2^n$ in general and $K \subseteq \sqrt{n}B_2^n$ for symmetric K , and show that the simplex and the cube make these bounds sharp.
 4. (classical) Prove that the ellipse of maximal area inscribed in a triangle is the Steiner inellipse (it touches the sides at their midpoints), and that its area equals $\frac{\pi}{3\sqrt{3}}$ times the area of the triangle. (*Hint: reduce to an equilateral triangle by an affine map.*)

5. (IMC) Let $B \subseteq \mathbb{R}^2$ be a bounded closed convex symmetric set with boundary Γ , and suppose the ellipse of maximal area inside B is the unit disc with boundary circle C . Prove that every arc of C of length at least $\pi/2$ meets Γ .